

HANDS-ON WORKSHOP · HERBSTSEMINAR 2026

Plausible Is Not Correct

Reliable LLM use in bioinformatics and cheminformatics research

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THE ONE IDEA

The LLM proposes. **A curated database decides.**

Every demo today turns an LLM answer into checkable claims and checks them against Crossref, PubMed, NCBI Gene, UniProt, PubChem, Wikidata or openFDA.

What a language model does

- 1** It continues text with **likely** next tokens, learned from its training data.
- 2** By default it does **not look anything up**.
- 3** Right and wrong answers come out **equally fluent**. Fluency is not evidence.

Two kinds of error

Factuality

The output conflicts with world knowledge.

“BRCA1 is on chromosome 13.” It is on 17;
BRCA2 is on 13.

Faithfulness

The output does not follow the given source or instruction.

A summary that adds a number the abstract does not contain.

Why it happens

Long-tail knowledge

Accuracy tracks how often related documents appeared in pre-training. Rare genes and compounds suffer most.

Kandpal et al. 2023, ICML

Guessing is rewarded

Training and benchmark scoring reward a guess over “I don’t know”.

Kalai et al. 2025

Sycophancy

Assistants tend to agree with the view stated in the prompt. Keep your hypothesis out of the question.

Sharma et al. 2024, ICLR

Snowballing

Models defend early errors, yet ChatGPT and GPT-4 recognised 67% and 87% of their own mistakes when asked separately.

Zhang et al. 2024, ICML

The evidence

Failure	Finding
Fabricated references	55% of GPT-3.5 and 18% of GPT-4 citations fabricated (Walters & Wilder 2023, Sci Rep)
...still in 2026	ChatGPT-5: 7.13% of 2,736 references fabricated; 49.34% fully accurate (Bernstein et al. 2026, Cureus)
Atomic facts	ChatGPT: 58% of atomic facts in biographies supported (FActScore; Min et al. 2023, EMNLP)
Invented packages	≥5.2% (commercial) and 21.7% (open-source) hallucinated packages; 205,474 unique fake names (Spracklen et al. 2025, USENIX Security)

WORKING PRINCIPLE

**Use LLMs where errors are cheap to detect,
and design your workflow so that errors
become cheap to detect.**

Are these references real?

Constructed reference	Verdict	Why
AlphaFold (Jumper 2021), correct DOI	VERIFIED	matches Crossref
BLAST (Altschul 1990), correct PMID	VERIFIED	matches PubMed
Gapped BLAST title with the 1990 BLAST DOI	MISMATCH	DOI of another paper
AlphaFold with year 2020	MISMATCH	record says 2021
Invented DOI 10.1038/s41586-023-99999-9	NOT_FOUND	does not resolve
Clustal Omega, title only	FOUND_BY_TITLE	PMID 21988835
Invented title, no identifier	UNVERIFIABLE	not proof of fake

Gene and protein facts

Claim	Database	Verdict
TP53 protein: 393 aa	UniProt P04637: 393	supported
CFTR protein: 1480 aa	UniProt P13569: 1480	supported
EGFR protein: 1210 aa	UniProt P00533: 1210	supported
BRCA1 on chromosome 13	NCBI Gene 672: 17q21.31	contradicted
HBB on chromosome 11	NCBI Gene 3043: 11p15.4	supported
DMD on chromosome Y	NCBI Gene 1756: Xp21.2-p21.1	contradicted

Ask for **JSON** and **identifiers**: verification becomes a loop, not a reading exercise.

Verifiers need verifying too

A UniProt search for the gene HTT returns two reviewed human entries:

Huntingtin

P42858 · official symbol HTT · 3,142 aa

Serotonin transporter

P31645 · official symbol SLC6A4 · synonym HTT ·
630 aa

Taking the first hit marks a correct answer as wrong. **Match on the official symbol or the accession.**

Five levers, cheapest first

Lever	What it means	Example
Ask better	identifiers, JSON, “null if unsure”, neutral wording	Demo 2 prompt
Ground	give the source text, demand exact quotes	RAG (Lewis et al. 2020); Demo 4
Use tools	let the model query databases	GeneGPT; ChemCrow
Measure uncertainty	sample several answers; disagreement is a warning	SelfCheckGPT; semantic entropy
Verify	check every claim against a curated source	this notebook

Lookup beats recall

0.12

ChatGPT, answering from memory

0.83

GeneGPT, calling NCBI Web APIs

Average score on the GeneTuring genomics benchmark. The gain comes from **querying a database** instead of recalling.

Reasoning helps, but it is not an explanation

Helps

Step-by-step prompting improves multi-step tasks.
Majority voting over several reasoning paths helps further.

Wei et al. 2022 · Wang et al. 2023

But

With inputs biased toward wrong answers, models wrote plausible rationales without mentioning the bias. Accuracy fell by up to 36%.

Turpin et al. 2023, NeurIPS

Treat written reasoning as a **draft argument to check**. Never accept “are you sure?” as verification.

Molecules: compare structures, not strings

Claimed structure	Result (InChIKey vs PubChem)
caffeine, paracetamol, ibuprofen	correct
“aspirin” written as the methyl ester	different molecule
levodopa with the wrong stereocentre (D-DOPA)	same skeleton, different stereo
<chem>C1=CC=CC=C1C(=O</chem>	invalid SMILES

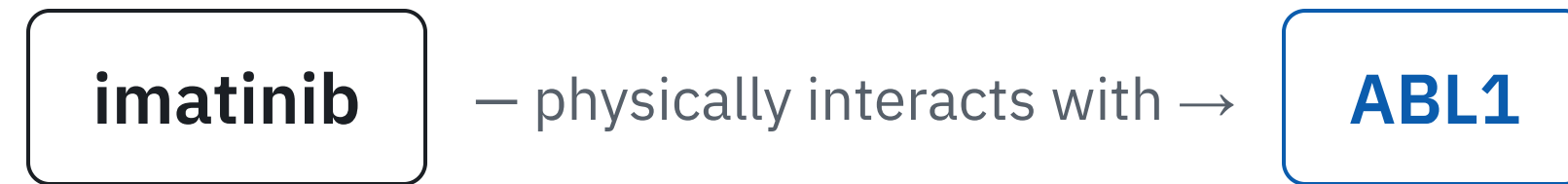
Ask for a **name or PubChem CID**, then look the structure up. Do not ask the model to write SMILES from memory.

Quote, or it didn't happen

Extracted item	Quote check
GeneGPT 0.83	EXACT
new Bing 0.44	EXACT
ChatGPT 0.21 (abstract says 0.12)	FUZZY 0.94: hidden change
GPT-4 0.71 (not in abstract)	ABSENT

A quote check proves the text exists, not that it supports the claim. It shrinks what you must read.

Facts as checkable triples



KG-enhanced LLMs

retrieve triples to ground or check answers

LLM-augmented KGs

extract candidate triples from text, then verify them

Synergised

both directions, iteratively

Imatinib: knowledge graph vs drug label

Claim	Wikidata	FDA label
interacts with ABL1	supported	—
interacts with EGFR (not a target)	not in KG	—
treats chronic myeloid leukemia	not in KG	listed
treats acute myeloid leukemia	in KG	not listed
treats gastrointestinal stromal tumor	in KG	listed

Three lessons

Absent $\neq$ false

Open-world assumption: “not in KG” mixes true and false claims.

Present $\neq$ true

Curated and crowd-sourced graphs contain errors.

Context matters

“Treats” depends on approval, population, source and version. Record provenance; triangulate.

Where do facts live, and can we fix them?

Knowledge neurons

Neurons whose activation tracks a fact; editing them updates or erases it. Dai et al. 2022, ACL

ROME

Factual recall localised to mid-layer MLPs; a rank-one edit changes one association. Meng et al. 2022, NeurIPS

Cross-lingual editing

Editing methods show clear limits when an edit must carry across languages, especially across script families. Beniwal et al., Findings of EACL 2024

Where does toxicity live?

Localised to layers and neurons, suppressed by scaling or rank-one edits; concentrated in early MLP layers. Beniwal & Singh 2026, preprint

Reliability is not language-neutral

17

PolyGuard

Multilingual safety moderation in 17 languages; 1.91M training and 29K test samples; +5.5% over existing classifiers.

Kumar, ..., Beniwal et al., COLM 2025

3×

Low-resource risk

Without any attack, low-resource languages were about 3× as likely to yield harmful output.

Deng et al. 2024, ICLR

38

Sycophancy

6 models, 1.1M instances, 38 languages: sycophancy rises sharply in low-resource languages.

Shah, Beniwal et al. 2026, preprint

Demo 6: ask the same question in English, German, Czech and Hindi, and compare with NCBI Gene.

Tools worth knowing

Literature

PubMed · Europe PMC · Semantic Scholar · PaperQA2 · Zotero

Structure

AlphaFold DB and Server · ColabFold · Boltz · Chai-1 · ESM · Foldseek

Design

ProteinMPNN · RFdiffusion · REINVENT 4

Omics

Biopython · Scanpy · scvi-tools · scGPT · Evo 2

Chemistry

RDKit · PubChem · ChEMBL · DeepChem · Chemprop · DiffDock

Private and local

Ollama · LM Studio · Hugging Face Hub

A prompt that makes answers checkable

Task: Extract all reported binding affinities from the abstract below.

Source: the abstract, pasted in full

Format: JSON with keys compound, target, value, unit, quote

Rules: quote copied exactly; null if a field is not stated

Exclude: nothing from memory or other sources

Also: ask for identifiers · use neutral wording · re-run important questions · log model, version, date and prompt.

Using AI safely in research

Data

Genetic and health data are special categories under GDPR Art. 9(1). No patient data, unpublished results or credentials in consumer chatbots; use local or institution-approved tools.

Authorship

AI tools are not authors. Humans are accountable for all content. Disclose which tools, for what, and to what extent.

Peer review

EU: refrain from substantial use. DFG (since 16 April 2026): supporting use only, in confidential systems, disclosed.

Integrity

Never use AI to fabricate, falsify or manipulate data. Read and test generated code; verify every package.

TAKE HOME

Seven rules

- 1** Ask for identifiers, not names.
- 2** Ask for JSON, then verify every field.
- 3** Demand exact quotes; read the near-matches.
- 4** Allow abstention; count it separately.
- 5** Sample more than once; agreement is not proof.
- 6** KG absence $\neq$ false; presence $\neq$ true.
- 7** Report AI use like an instrument.

Reporting AI use: an illustrative methods statement

“We used Claude Opus 5.5 (Anthropic; API; 1–14 October 2026) to extract gene–disease associations from 120 PubMed abstracts. Prompts are provided in Supplementary File S2. Every extracted association was checked for an exact supporting quote and against NCBI Gene (accessed 15 October 2026); 9 of 412 items were corrected and 14 removed. The authors take full responsibility for the content.”

Name the model and version, dates, task, where the prompts are, what was verified against which source, and what changed.

EVERYTHING FROM TODAY

`github.com/himanshubeniwal/ plausible-is-not-correct`

Notebook: runs in Google Colab, no install

Cheat sheet: rules and prompt templates

Workshop notes with all demo results

AI research guidelines: data, authorship, review

Curated tools for bio- and cheminformatics

Verified references with DOIs

Thank you · Himanshu Beniwal · ScaDS.AI / TU Dresden